

I U S

Faculté des Sciences et des Technologies

Administration des Systèmes Informatiques I

TD n° 9

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Mise en place d'un serveur mail d'entreprise(Postfix + Dovecot +Roundcube)

- installer un serveur SMTP
- configurer la réception IMAP / POP3
- déployer un webmail professionnel
- sécuriser les accès
- analyser les logs
- tester l'envoi / réception
- documenter une architecture mail complète

Étape 1 — Planification du domaine et des utilisateurs

Domaine choisi : ius.edu.ht **Hostname :** mail.ius.edu.ht **IP serveur :** 192.168.56.10 **OS :** Ubuntu Server 24.04

liste des 40 utilisateurs

25 Étudiants — format etudiantNOM@ius.edu.ht

etudiant-anne@ius.edu.ht, etudiant-alain@ius.edu.ht, etudiant-brian@ius.edu.ht, etudiant-christy@ius.edu.ht, etudiant-cliff@ius.edu.ht, etudiant-donsam@ius.edu.ht, etudiant-emile@ius.edu.ht, etudiant-evena@ius.edu.ht, etudiant-frantzdy@ius.edu.ht, etudiant-gabard@ius.edu.ht, etudiant-henry@ius.edu.ht, etudiant-julien@ius.edu.ht, etudiant-lens@ius.edu.ht, etudiant-lyne@ius.edu.ht, etudiant-magdala@ius.edu.ht, etudiant-narwens@ius.edu.ht, etudiant-oliver@ius.edu.ht, etudiant-pierre@ius.edu.ht, etudiant-ryan@ius.edu.ht, etudiant-ruth@ius.edu.ht, etudiant-rose@ius.edu.ht, etudiant-youri@ius.edu.ht, etudiant-tracy@ius.edu.ht, etudiant-tobias@ius.edu.ht

10 Professeurs — format prof.NOM@ius.edu.ht

prof.ismael@ius.edu.ht, prof.lorens@ius.edu.ht, prof.lamour@ius.edu.ht, prof.jetho@ius.edu.ht, prof.jude@ius.edu.ht, prof.beslin@ius.edu.ht, prof.casimir@ius.edu.ht, prof.etienne@ius.edu.ht, prof.wiycleph@ius.edu.ht, prof.sophie@ius.edu.ht

5 Administratifs — format admin.NOM@ius.edu.ht

admin.zidor@ius.edu.ht, admin.ketsia@ius.edu.ht, admin.fred@ius.edu.ht, admin.providence@ius.edu.ht, admin.sylvestre@ius.edu.ht

Étape 2 — Préparation du système

```
christy@christy:~$ sudo hostnamectl set-hostname mail.ius.edu.ht
[sudo] password for christy:
christy@christy:~$ sudo nano /etc/hosts
christy@christy:~$ hostname -f
mail.ius.edu.ht
christy@christy:~$ |
```

Étape 3 — Installation de Postfix (SMTP)

```
christy@mail:~$ sudo apt update
[sudo] password for christy:
Atteint :1 http://us.archive.ubuntu.com/ubuntu noble InRelease
Atteint :2 http://security.ubuntu.com/ubuntu noble-security InRelease
Atteint :3 https://packages.grafana.com/oss/deb stable InRelease
Atteint :4 http://us.archive.ubuntu.com/ubuntu noble-updates InRelease
```

```
christy@mail:~$ sudo apt install postfix mailutils -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
postfix est déjà la version la plus récente (3.8.6-1build2).
postfix passé en « installé manuellement ».
Les paquets supplémentaires suivants seront installés :
  gsasl-common guile-3.0-libs libgc1 libgsasl18 libgssglue1 libidn12
  libltdl7 libmailutils9t64 libmysqlclient21 libntlm0 libpq5
  mailutils-common mysql-common
```

```
christy@mail:~$ cat /etc/postfix/main.cf | grep myorigin
#myorigin = /etc/mailname
christy@mail:~$ sudo nano /etc/postfix/main.cf
christy@mail:~$ |
```

```
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated
myhostname = mail.ius.edu.ht
mydomain = ius.edu.ht
myorigin = $mydomain
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
mydestination = $myhostname, christy, localhost.localdomain, $mydomain,
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128, 192.168.56.0
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all
home_mailbox = Maildir /
```

```

christy@mail:~$ sudo nano /etc/postfix/main.cf
christy@mail:~$ sudo nano /etc/postfix/main.cf
christy@mail:~$ sudo systemctl restart postfix
Command 'suod' not found, did you mean:
  command 'sudo' from deb sudo (1.9.15p5-3ubuntu5.24.04.2)
  command 'sudo' from deb sudo-ldap (1.9.15p5-3ubuntu5.24.04.2)
Try: sudo apt install <deb name>
christy@mail:~$ sudo systemctl restart postfix
christy@mail:~$ sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; ▾
   Active: active (exited) since Sat 2026-05-02 15:45:54 UTC; 12s ago
     Docs: man:postfix(1)
   Process: 323995 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
    Main PID: 323995 (code=exited, status=0/SUCCESS)
      CPU: 58ms

mai 02 15:45:54 mail.ius.edu.ht systemd[1]: Starting postfix.service - ▾
mai 02 15:45:54 mail.ius.edu.ht systemd[1]: Finished postfix.service - ▾
lines 1-10/10 (END)
christy@mail:~$

```

Étape 4 — Installation de Dovecot (IMAP/POP3)

```

christy@mail:~$ sudo apt install dovecot-imapd dovecot-pop3d -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  dovecot-core libexttextcat-2.0-0 libexttextcat-data
Paquets suggérés :
  dovecot-gssapi dovecot-ldap dovecot-lmtpd dovecot-managesieved
  dovecot-mysql dovecot-pgsql dovecot-sieve dovecot-solr
  dovecot-sqlite dovecot-submissiond ntp
Les NOUVEAUX paquets suivants seront installés :
  dovecot-core dovecot-imapd dovecot-pop3d libexttextcat-2.0-0
  libexttextcat-data

```

```

christy@mail:~$ sudo nano /etc/dovecot/conf.d/10-mail.conf
christy@mail:~$

```

```

#
mail_driver = maildir
mail_path = ~/Maildir

```

```

christy@mail:~$ sudo doveconf -n
# 2.3.21 (47349e2482): /etc/dovecot/dovecot.conf
# Pigeonhole version 0.5.21 (f6cd4b8e)
doveconf: Fatal: Error in configuration file /etc/dovecot/conf.d/10-mail
.conf line 30: Unknown setting: mail_driver

```

Active les protocoles IMAP et POP3 :

```

christy@mail:~$ sudo nano /etc/dovecot/dovecot.conf
christy@mail:~$

```

```

christy@mail:~$ sudo systemctl restart dovecot
christy@mail:~$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; >
   Active: active (running) since Sat 2026-05-02 16:10:57 UTC; 5s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
  Main PID: 327707 (dovecot)
    Status: "v2.3.21 (47349e2482) running"
     Tasks: 4 (limit: 4601)
    Memory: 3.4M (peak: 3.5M)
       CPU: 205ms
    CGroup: /system.slice/dovecot.service
            └─327707 /usr/sbin/dovecot -F
              └─327708 dovecot/anvil
                └─327709 dovecot/log
                  └─327710 dovecot/config

mai 02 16:10:57 mail.ius.edu.ht systemd[1]: Starting dovecot.service - >
mai 02 16:10:57 mail.ius.edu.ht dovecot[327707]: master: Dovecot v2.3.2>
mai 02 16:10:57 mail.ius.edu.ht systemd[1]: Started dovecot.service - D>

christy@mail:~$ |

```

```

GNU nano 7.2 /etc/dovecot/dovecot.conf *
# source/destination IPs by placing the settings inside sections, for e>
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }
protocols = imap pop3|

```

Étape 5 — Création des utilisateurs

```

christy@mail:~$ sudo adduser anne
info: Adding user 'anne' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'anne' (1003) ...
info: Adding new user 'anne' (1003) with group 'anne (1003)' ...
info: Creating home directory '/home/anne' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for anne
Enter the new value, or press ENTER for the default

christy@mail:~$ for user in anne alain brian gerys cliff donsam emile ev
ena frantzdy gabard henry julien lens lyne magdala narwens oliver pierre
ryan ruth rose youri tracy tobias abimael; do
  sudo -u $user mkdir -p /home/$user/Maildir/{cur,new,tmp}
done

christy@mail:~$ sudo ls /home/anne/Maildir
cur new tmp
christy@mail:~$

```

Étape 6 — Création des professeurs

```
christy@mail:~$ for user in ismael lorens lamour jetho jude etienne besl  
in wiycleph sophie casimir; do  
    sudo -u $user mkdir -p /home/$user/Maildir/{cur,new,tmp}  
done  
christy@mail:~$ sudo ls /home/ismael/Maildir  
cur new tmp  
christy@mail:~$ sudo ls /home/casimir/Maildir  
cur new tmp  
christy@mail:~$
```

Étape 7 — Création des administrateurs

```
christy@mail:~$ for user in zidor ketsia fred providence sylvestre; do  
    sudo -u $user mkdir -p /home/$user/Maildir/{cur,new,tmp}  
done  
christy@mail:~$ sudo ls /home/zidor/Maildir  
sudo ls /home/sylvestre/Maildir  
cur new tmp  
cur new tmp  
christy@mail:~$
```

Étape 8 — Installation de Roundcube (Webmail)

Installe Apache + PHP :

```
christy@mail:~$ sudo systemctl start apache2  
christy@mail:~$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled;   
   Active: active (running) since Sat 2026-05-02 16:53:30 UTC; 2s ago  
     Docs: https://httpd.apache.org/docs/2.4/  
  Process: 340640 ExecStart=/usr/sbin/apachectl start (code=exited, s  
 Main PID: 340643 (apache2)  
    Tasks: 6 (limit: 4601)  
  Memory: 14.7M (peak: 15.0M)  
     CPU: 424ms  
   CGroup: /system.slice/apache2.service  
           └─340643 /usr/sbin/apache2 -k start  
             └─340645 /usr/sbin/apache2 -k start  
               └─340646 /usr/sbin/apache2 -k start  
                 └─340647 /usr/sbin/apache2 -k start  
                   └─340648 /usr/sbin/apache2 -k start  
                     └─340649 /usr/sbin/apache2 -k start  
  
mai 02 16:53:30 mail.ius.edu.ht systemd[1]: Starting apache2.service -   
mai 02 16:53:30 mail.ius.edu.ht systemd[1]: Started apache2.service - T  
lines 1-19/19 (END)  
christy@mail:~$
```

```
christy@mail:~$ sudo apt install mysql-server -y  
Lecture des listes de paquets... Fait  
Construction de l'arbre des dépendances... Fait  
Lecture des informations d'état... Fait  
Les paquets suivants ont été installés automatiquement et ne sont plus n  
écessaires :
```



```
christy@mail:~$ cat /etc/aliases
# See man 5 aliases for format
postmaster:    root
etudiants:     anne,alain,brian,gerys,cliff,donsam,emile,evena,frantzdy,gabar
d,henry,julien,lens,lyne,magdale,narwens,oliver,pierre,ryan,ruth,rose,you
ri,tracy,tobias,abimael

professeurs:   ismael,lorens,lamour,jetho,jude,etienne,beslin,wiycleph,soph
ie,casimir

administration: zidor,ketsia,fred,providence,sylvestre

tous:          etudiants,professeurs,administration
christy@mail:~$
```

Étape 9 — Tests fonctionnels

```
christy@mail:~$ echo "Bonjour professeur Ismael, j'ai une question sur le
cours" | mail -s "Question de cours" ismael
christy@mail:~$ echo "Bonjour Anne, voici la réponse à ta question" | mai
l -s "Réponse du professeur" anne
christy@mail:~$ echo "Réunion générale demain à 9h" | mail -s "Annonce im
portante" tous
christy@mail:~$
```



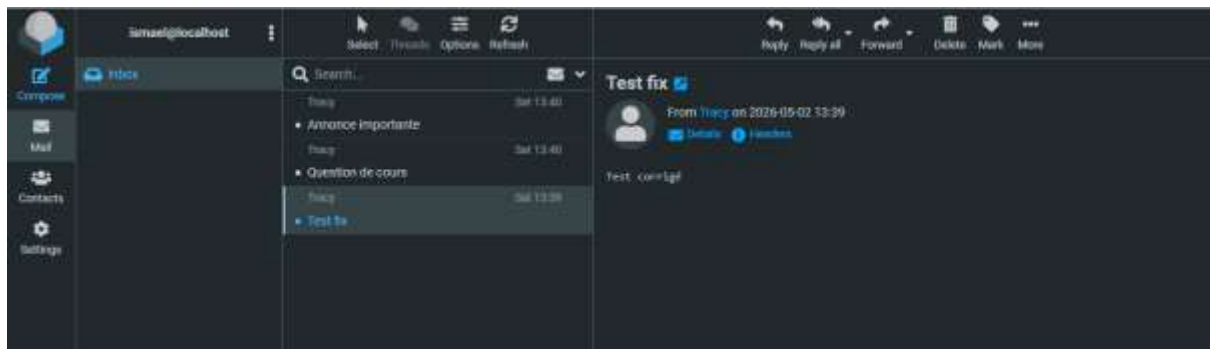
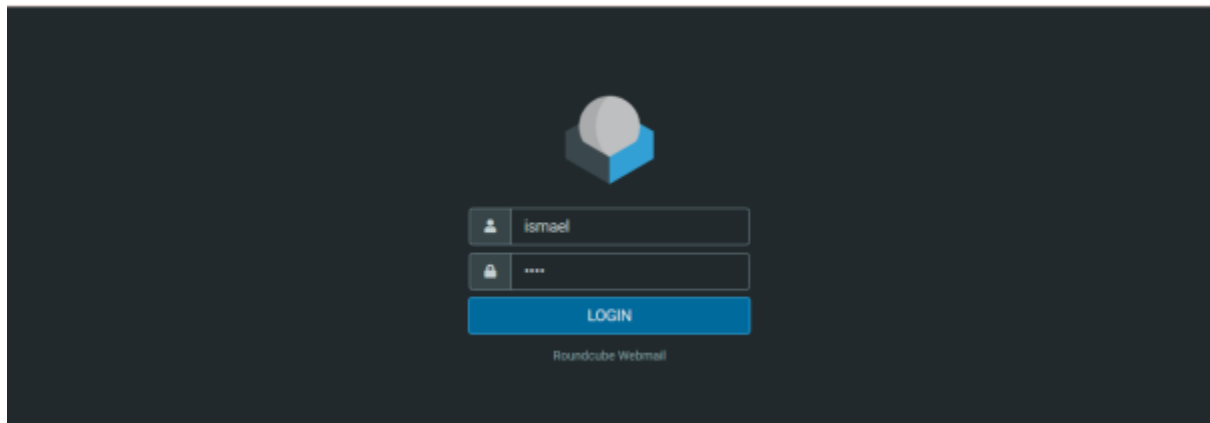

```
christy@mail:~$ sudo ufw status
Status: active
```

To	Action	From
--	-----	----
2222/tcp	ALLOW	Anywhere
443/tcp	ALLOW	Anywhere
80/tcp	ALLOW	Anywhere
9090/tcp	ALLOW	Anywhere
9100/tcp	ALLOW	Anywhere
3000/tcp	ALLOW	Anywhere
25	ALLOW	Anywhere
143	ALLOW	Anywhere
993	ALLOW	Anywhere
80	ALLOW	Anywhere
2222/tcp (v6)	ALLOW	Anywhere (v6)
443/tcp (v6)	ALLOW	Anywhere (v6)
80/tcp (v6)	ALLOW	Anywhere (v6)
9090/tcp (v6)	ALLOW	Anywhere (v6)
9100/tcp (v6)	ALLOW	Anywhere (v6)
3000/tcp (v6)	ALLOW	Anywhere (v6)
25 (v6)	ALLOW	Anywhere (v6)
143 (v6)	ALLOW	Anywhere (v6)
993 (v6)	ALLOW	Anywhere (v6)
80 (v6)	ALLOW	Anywhere (v6)

```
christy@mail:~$ |
```

```
christy@mail:~$ echo "Test logs" | mail -s "Test analyse logs" anne
christy@mail:~$
```

```
christy@mail:~$ sudo tail -f /var/log/mail.log
2026-05-04T03:09:28.701554+00:00 christy dovecot: master: Warning: Time
moved forwards by 120341.767754 seconds - adjusting timeouts.
2026-05-04T03:11:06.033761+00:00 christy dovecot: master: Warning: Time
moved forwards by 0.154139 seconds - adjusting timeouts.
2026-05-04T03:15:49.749947+00:00 christy postfix/pickup[351350]: B541011
E: uid=1000 from=<christy>
2026-05-04T03:15:49.782820+00:00 christy postfix/cleanup[405956]: B54101
1E: message-id=<20260504031549.B541011E@mail.ius.edu.ht>
2026-05-04T03:15:49.795563+00:00 christy postfix/qmgr[350336]: B541011E:
from=<christy@ius.edu.ht>, size=402, nrcpt=1 (queue active)
2026-05-04T03:15:49.830904+00:00 christy postfix/local[405958]: B541011E
: to=<anne@ius.edu.ht>, orig_to=<anne>, relay=local, delay=0.13, delays=
0.1/0.03/0/0.01, dsn=2.0.0, status=sent (delivered to maildir)
2026-05-04T03:15:49.831705+00:00 christy postfix/qmgr[350336]: B541011E:
removed
```



Professeurs — format prof.NOM@ius.edu.ht **5 Administratifs** — format
admin.NOM@ius.edu.ht